

Edexcel International A Level

Mathematics & Further Mathematics — Content Index (Y12 / Y13)

A chapter-level index of the Edexcel (Pearson) **International Advanced Level (IAL)** in Mathematics and Further Mathematics (2018 specification), aligned to the Pearson IAL student-book series used at international schools such as Cranleigh Abu Dhabi.

A Level Mathematics = P1, P2, P3, P4 + two applied units (here M1, S1). • **A Level Further Mathematics** = FP1, FP2, FP3 + applied units (here D1, M2, S2).

Year split (typical). Y12 (International AS) usually covers P1, P2 and one applied unit; Y13 (International A2) covers P3, P4 and a second applied unit. For Further Maths, FP1 and lower-numbered applied units sit in Y12, with FP2/FP3 and higher applied units in Y13. Exact sequencing varies by centre.

Accuracy note. P1 is verified chapter-by-chapter against the actual book contents page. For the other units, chapter titles follow the Pearson IAL student books and the bullets list the key content within each chapter — spot-check the fine detail against each book's contents page.

Part 1 — Mathematics ("normal maths")

P1 — Pure Mathematics 1 VERIFIED

1. **Algebraic expressions** — index laws; expanding brackets; factorising; negative and fractional indices; surds; rationalising denominators
2. **Quadratics** — solving quadratic equations; completing the square; functions; quadratic graphs; the discriminant
3. **Equations and inequalities** — linear & quadratic simultaneous equations; simultaneous equations on graphs; linear & quadratic inequalities; inequalities on graphs; regions
4. **Graphs and transformations** — cubic graphs; reciprocal graphs; points of intersection; translating graphs; stretching graphs; transforming functions
5. **Straight line graphs** — $y = mx + c$; equations of straight lines; parallel and perpendicular lines; length and area
6. **Trigonometric ratios** — cosine rule; sine rule; areas of triangles; solving triangle problems; graphs of $\sin$, $\cos$, $\tan$; transforming trig graphs
7. **Radians** — radian measure; arc length; areas of sectors and segments
8. **Differentiation** — gradients of curves; finding the derivative; differentiating x^n and quadratics; multi-term functions; tangents and normals; second order derivatives
9. **Integration** — integrating x^n ; indefinite integrals; finding functions

P2 — Pure Mathematics 2

1. **Algebraic methods** — proof (deduction, exhaustion, counter-example); dividing polynomials; the factor theorem
2. **Coordinate geometry in the (x, y) plane** — midpoints and perpendicular bisectors; equation of a circle; intersections of lines and circles; tangent and chord properties; circles and triangles
3. **Exponentials and logarithms** — exponential functions; $y = e^x$; exponential modelling; logarithms; laws of logs; solving equations with logs; natural logarithms; non-linear data
4. **The binomial expansion** — Pascal's triangle; factorial notation; the binomial expansion; solving problems; binomial estimation
5. **Sequences and series** — arithmetic and geometric sequences and series; sum to infinity; sigma notation; recurrence relations; modelling
6. **Trigonometric identities and equations** — angles in four quadrants; exact values; trig identities; simple and harder trig equations
7. **Differentiation** — increasing/decreasing functions; stationary points; sketching gradient functions; modelling
8. **Integration** — definite integrals; area under a curve; area between curves and lines; the trapezium rule

P3 — Pure Mathematics 3

1. **Algebraic methods** — partial fractions (proper and improper); algebraic division
2. **Functions and graphs** — the modulus function; functions and mappings; composite functions; inverse functions; transformations $y = |f(x)|$ and $y = f(|x|)$
3. **Trigonometric functions** — secant, cosecant, cotangent; identities; inverse trigonometric functions
4. **Trigonometric addition formulae** — addition and double-angle formulae; the $R \cos(\theta \pm \alpha)$ form; solving equations; proving identities
5. **Exponentials and logarithms** — e^x and $\ln x$; derivatives; exponential modelling, growth and decay
6. **Differentiation** — chain, product and quotient rules; e^x , $\ln x$ and trig functions; implicit and parametric differentiation; rates of change
7. **Integration** — standard integrals; substitution; by parts; using partial fractions; areas; trapezium rule
8. **Numerical methods** — locating roots; iteration; the Newton–Raphson method

P4 — Pure Mathematics 4

1. **Proof** — proof by contradiction
2. **Partial fractions** — into partial fractions; improper fractions; applications
3. **Coordinate geometry in the (x, y) plane** — parametric equations; converting to Cartesian; parametric curves; points of intersection
4. **Binomial expansion** — expansion for any rational index; using partial fractions; approximations and validity
5. **Differentiation** — parametric differentiation; connected rates of change; forming differential equations
6. **Integration** — further techniques; volumes of revolution; solving differential equations (separating variables)
7. **Vectors** — 3D coordinates; vectors in three dimensions; equation of a line in 3D; points of intersection; the scalar (dot) product; angles

M1 — Mechanics 1

1. **Mathematical models in mechanics** — modelling assumptions; quantities and S.I. units
2. **Constant acceleration** — displacement–time and velocity–time graphs; the suvat formulae; vertical motion under gravity
3. **Vectors in mechanics** — i, j notation; velocity, acceleration and force as vectors
4. **Dynamics of a particle moving in a straight line** — force diagrams; Newton's laws; motion in two dimensions; connected particles and pulleys
5. **Forces and friction** — resolving forces; inclined planes; friction and the coefficient of friction
6. **Momentum and impulse** — momentum and impulse; conservation of linear momentum; direct impact/collisions
7. **Statics of a particle** — equilibrium of a particle; forces in equilibrium; statics with friction
8. **Moments** — moment of a force; resultant moments; equilibrium of rigid bodies; tilting and non-uniform rods

S1 — Statistics 1

1. **Mathematical models in probability and statistics** — the modelling process
2. **Measures of location and spread** — mean, median, mode; quartiles and percentiles; variance and standard deviation; coding
3. **Representation of data** — histograms; box plots; outliers; skewness; comparing distributions
4. **Probability** — sample spaces; Venn diagrams; mutually exclusive and independent events; conditional probability; tree diagrams
5. **Correlation** — scatter diagrams; the product moment correlation coefficient; interpretation
6. **Regression** — the least-squares regression line; interpolation and extrapolation
7. **Discrete random variables** — probability distributions; the cumulative distribution function; $E(X)$; $\text{Var}(X)$
8. **The Normal distribution** — the standard normal distribution; finding probabilities; finding unknown means/standard deviations

Part 2 — Further Mathematics

FP1 — Further Pure Mathematics 1

1. **Complex numbers** — imaginary and complex numbers; complex arithmetic; conjugates; Argand diagrams; modulus and argument; modulus–argument form
2. **Roots of quadratic equations** — roots with real/complex coefficients; sums and products of roots; forming equations
3. **Numerical solutions of equations** — locating roots; interval bisection; linear interpolation; Newton–Raphson
4. **Coordinate systems** — the parabola and rectangular hyperbola; parametric forms; tangents and normals; loci
5. **Matrices** — matrix operations; multiplication; determinants; inverse of a 2×2 matrix
6. **Transformations using matrices** — linear transformations; reflections and rotations; enlargements and stretches; successive transformations
7. **Series** — standard summation results (Σr , Σr^2 , Σr^3); summing series
8. **Proof by induction** — mathematical induction; divisibility; recurrence relations; matrix statements

FP2 — Further Pure Mathematics 2

1. **Inequalities** — algebraic inequalities; modulus inequalities; graphical methods
2. **Series** — the method of differences
3. **Further complex numbers** — de Moivre's theorem; trig identities; sums of series; n th roots; loci and regions; exponential form $re^{i\theta}$
4. **First order differential equations** — separating variables; exact equations; the integrating factor; first-order linear; reducible equations
5. **Second order differential equations** — homogeneous (auxiliary equation); non-homogeneous (particular integral); boundary conditions; modelling
6. **Maclaurin and Taylor series** — Maclaurin series; Taylor series; series solutions of differential equations
7. **Polar coordinates** — polar and Cartesian coordinates; sketching curves; area enclosed by a polar curve; tangents

FP3 — Further Pure Mathematics 3

1. **Hyperbolic functions** — definitions and graphs; inverse hyperbolic functions; identities; solving equations
2. **Further coordinate systems** — the ellipse and hyperbola; eccentricity; foci and directrices; tangents and normals; loci
3. **Differentiation** — differentiating inverse trigonometric and hyperbolic functions
4. **Integration** — inverse trig/hyperbolic forms; standard results; reduction formulae; arc length; surface of revolution
5. **Vectors** — the vector (cross) product; scalar triple product; equations of lines and planes; distances and angles
6. **Further matrix algebra** — 3×3 determinants and inverses; systems of linear equations; eigenvalues and eigenvectors; diagonalisation

D1 — Decision Mathematics 1

1. **Algorithms** — using and understanding algorithms; flow charts; sorting (bubble, quick sort); bin-packing; order of an algorithm
2. **Graphs and networks** — modelling with graphs; terminology; special graphs; adjacency matrices
3. **Algorithms on graphs** — minimum spanning trees (Kruskal's, Prim's); Dijkstra's shortest-path algorithm
4. **Route inspection** — the route inspection (Chinese postman) problem; Eulerian graphs
5. **Critical path analysis** — precedence tables; activity networks; dummy activities; early/late event times; critical activities and float; Gantt charts; scheduling
6. **Linear programming** — formulating LP problems; graphical solutions; objective lines and the vertex method; integer solutions
7. **Matchings** — bipartite graphs; matchings; alternating paths; the maximum matching algorithm

M2 — Mechanics 2

1. **Kinematics** — variable acceleration; calculus for kinematics; motion in two dimensions; projectiles (horizontal/vertical components, projectile formulae)
2. **Centres of mass** — systems of particles; uniform laminae and standard shapes; frameworks; equilibrium of a suspended lamina
3. **Work, energy and power** — work done by a force; kinetic and potential energy; the work–energy principle; power
4. **Collisions** — momentum and impulse; conservation of momentum; Newton's law of restitution; direct impact; impact with a surface; successive impacts
5. **Statics of rigid bodies** — moments and resultant moments; equilibrium of rigid bodies; friction

S2 — Statistics 2

1. **The binomial distribution** — the binomial model; using tables; mean and variance
2. **The Poisson distribution** — the Poisson model; using tables; mean and variance; Poisson approximation to the binomial
3. **Continuous random variables** — the probability density function; the cumulative distribution function; mean, variance, mode and median
4. **Continuous distributions** — the continuous uniform (rectangular) distribution; normal approximation to the binomial and Poisson
5. **Hypothesis testing** — populations and sampling; test statistics; tests for a binomial proportion and a Poisson mean; critical regions and significance levels

Unit-to-qualification map

Qualification	Pure units	Applied units
A Level Mathematics	P1, P2, P3, P4	M1, S1
A Level Further Mathematics	FP1, FP2, FP3	D1, M2, S2

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